

ZIPING LIU

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SUMMARY

Full-stack Software Developer with 3 years of enterprise SaaS experience building production web applications for data-heavy business workflows. I translate ambiguous requirements into scalable, production-ready solutions and iterate quickly with product, design, QA, and engineering stakeholders. CS + Statistics background with current Applied Data Science master's training; available for 4- or 8-month co-op placements.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS

Frontend: React, Redux, GraphQL, AG Grid, Jest, responsive web UI

Backend/Cloud: Node.js, REST APIs, Flask, PostgreSQL, DynamoDB, Docker, AWS Lambda, AWS ECS, AWS S3, AWS EC2

Data/ML: Pandas, NumPy, scikit-learn, TensorFlow/Keras, Power BI, Jupyter/Colab

Tools: Git, GitHub Actions, VS Code, Postman, Linux/Unix

PROFESSIONAL EXPERIENCE

Software Developer | Varicent Canada Opco Ltd., Toronto, ON Oct 2021 - Oct 2024

- Delivered production full-stack features for an enterprise SaaS sales planning platform used by large-scale business clients, supporting quota planning and data-heavy workflows.
- Built and standardized quota planning screens and data grids with React, TypeScript, Redux, GraphQL, and AG Grid, including tabs, hierarchy drill-downs, pagination, filtering, sorting, resizing, and custom cells.
- Built frontend and backend data access logic using GraphQL, AWS Lambda, and SQL queries against PostgreSQL to enforce user-specific visibility and edit permissions.
- Served as feature lead for a full-stack initiative, scoping PM/design requirements, breaking down tasks, coordinating implementation discussions, and communicating timeline risks.

RECENT PROJECTS

ClothSnap | Machine Learning Operations, University of Ottawa Dec 2025

- Built an end-to-end MLOps pipeline for a 10-class clothing image classifier, achieving 87.1% test accuracy with TensorFlow/Keras and MobileNetV2.
- Deployed a Dockerized Flask prediction API with TensorFlow Serving on AWS EC2.
- Implemented GitHub Actions CI/CD to build and push images to GHCR, deploy to EC2, and monitor predictions and infrastructure health with AWS CloudWatch.

Car Recommendation System | Fundamentals of Applied Data Science, University of Ottawa Nov 2025

- Owned an unsupervised clustering pipeline for car recommendations using a cleaned 1,213-row vehicle dataset.
- Applied K-Means separately to EV and non-EV vehicles, selecting optimal K with Elbow and Silhouette diagnostics and visualizing clusters with PCA.
- Integrated clustering outputs into the downstream recommendation pipeline and helped package the final project with a clear repository structure and README.

EDUCATION

University of Ottawa, Ottawa, ON Sep 2025 - Sep 2027 expected

Master of Digital Transformation and Innovation, Concentration in Applied Data Science

- Cumulative GPA: A+ | Focused on applied data science, analytics, and data-driven decision-making, with coursework in machine learning, artificial intelligence, and data analytics applications.

University of Waterloo, Waterloo, ON Jan 2016 - Feb 2020

Bachelor of Computer Science, Statistics Minor

- Cumulative GPA: 83.3/100 | Award: University of Waterloo Merit Scholarship
- Relevant Courses: Object-Oriented Software Development, Data Structures, Computer Networks, Operating Systems, Database Management, Artificial Intelligence, Machine Learning